Project identity and purpose

- InfluenceIQ: an offline-first meeting-intelligence and review system.
- Meet IQ is the application label used in parts of the interface.
- Timestamped speech, speakers, optional slide context and source-linked candidates.
- A reviewer checks the outputs against the original recording.

Problem statement and intended users

- Long recordings take time to review and decisions need surrounding context.
- Speaking the longest does not necessarily mean contributing most to an outcome.
- A summary alone does not expose every speaker, timestamp or reasoning step.
- Intended use: review assistance for teams, students and meeting participants.

What is our contribution and novelty?

- An implemented local workflow combining audio, document context and source traceability.
- Structured candidate decisions, interactions and interpretable contribution features.
- Learned English discussion-review model alongside the rule-based workflow.
- Controlled fusion comparisons and exploratory outcome-alignment diagnostics.
- Established models are reused; no claim of a newly invented foundation model.

We do have a summary: why no generative notes?

- Implemented: structured executive summary from detected events and decisions.
- It includes proposals, candidate confirmed decisions and action-item excerpts.
- Not deployed: LLM-written abstractive notes or a meeting chatbot.
- Core requirements are traceability, local execution and reviewable outputs.
- Generative notes could be an optional future view, with factuality evaluation.

Modalities: what information enters the system?

- Audio: waveform, speech content, word times and speaker-related acoustic information.
- Optional PPTX documents: native text, tables, notes and document metadata.
- Downstream fusion compares ASR text with extracted document text.
- Slide-display timing is optional metadata; a PPTX file does not supply it automatically.
- No face, gaze, emotion, gesture or general chart-understanding model.

Complete system architecture

- Browser record/upload → authenticated FastAPI validation and queue.
- FFmpeg normalization → Whisper transcript → speaker diarization and word attribution.
- Optional PPT extraction → passage-to-slide alignment.
- Events/evidence → decision candidates → interactions and contribution scores.
- Report, history, audio playback, human corrections and exports.

Phase 1: input handling and audio preparation

- Accept ordered parts of one recording and optional ordered PPTX files.
- Validate type, decodability, options and aggregate upload size.
- FFmpeg converts audio to 16 kHz, mono, PCM16 WAV.
- Save inputs, job state and checkpoints for recovery.
- Concatenating recording parts is not microphone-array fusion or source separation.

Phase 2: speech recognition model

- Pretrained multilingual Whisper through faster-whisper / CTranslate2.
- Shared default: small model, CPU, int8, beam size 5, four threads.
- 120-second central chunks, word timestamps and speech filtering.
- Draft recognition appears before final whole-meeting speaker attribution.
- Pretrained weights are reused; Whisper was not trained by our team.

Phase 3: speaker diarization and word attribution

- Sherpa-ONNX: pretrained segmentation, speaker embeddings and whole-meeting clustering.
- Configured clustering threshold 0.9; optional exact known speaker count.
- Assign each word using greatest qualifying speaker-time overlap.
- Ties or inadequate coverage produce UNKNOWN; names remain reviewable labels.
- Diarization groups voices; it does not verify identity or separate mixed voices.

Phase 4: document extraction and semantic representation

- python-pptx extracts title/body text, tables and speaker notes.
- all-MiniLM-L6-v2 represents slide and transcript text as 384-dimensional vectors.
- Cosine similarity supplies the semantic matching score.
- TF-IDF is an implemented lexical alternative, not an extra averaged score.
- Native text extraction does not equal general visual understanding.

Which fusion techniques did we actually use?

- Temporal association: recognized words + speaker intervals.
- Alignment score fusion: semantic similarity + available slide-display overlap.
- Unified records: passage, speaker, times, slide candidates and scores.
- Discussion detector: concatenate word, character and context TF-IDF features.
- Discussion detector: weighted probability ensemble of forest and logistic classifiers.

Temporal–semantic alignment formula and fallback

- Semantic score: cosine between passage and slide embeddings.
- Temporal score: overlap with the slide's active interval / passage duration.
- Combined score = (0.5 × temporal + 0.5 × semantic) / (0.5 + 0.5).
- No valid slide timing → semantic-only; top 3 candidates, threshold 0.3.
- Below threshold → NO_CONFIDENT_SLIDE; similarity is not a probability.

Why these methods, and why not other fusion methods?

- Modular scores are inspectable and support missing document/timing inputs.
- Pure timing struggles with missing timestamps and references to earlier slides.
- Pure semantics struggles with vague phrases such as “this figure.”
- No jointly trained early-fusion or cross-attention model was implemented.
- Alternatives need aligned labels, computation and fair validation; no superiority claim.

Phase 5: events, evidence and decision candidates

- Default bilingual rules identify 11 event categories.
- Categories include proposals, evidence, objections, revisions, decisions and actions.
- Evidence relations use event type, speaker, timing and available slide links.
- Decision reconstruction groups related candidates and records source lineage.
- Detected “confirmed” is a system status, not human-verified correctness.

Separate learned discussion detector: what we trained

- English discussion membership: suggest passages for review, not final decisions.
- Word unigrams/bigrams + character 3–5 grams + up to ±40 context units.
- Concatenate sparse TF-IDF features; forest 75% + logistic regression 25%.
- Smooth over ±5 units; decision threshold approximately 0.35774.
- AMI split: 29 training, 6 development and 12 historical meetings.

Phase 6: interactions and participant contribution scores

- Build directed source-linked interactions from evidence and decision lineage.
- Seven contribution features; normalize each across meeting participants.
- Weights: proposal .15, evidence .20, objection .15, revision .20.
- Decision .15, interaction .05, slide grounding .10.
- A reviewable prototype score; not PageRank, causality or a personal rating.

Frontend, backend, storage and local operation

- React/Vite: record, upload, progress, transcript, review, charts and exports.
- FastAPI/Uvicorn: validation, accounts, queue, processing and private API access.
- SQLite/SQLAlchemy plus private audio, checkpoints and report files.
- bcrypt password hashing and JWT login; one serialized ML worker.
- Offline inference after model provisioning; cloud clients still need connectivity.

Outputs and human review

- Timestamped transcript, anonymous speaker groups and source audio playback.
- Optional slide candidates and alignment scores.
- Discussion passages, candidate decisions, actions and source references.
- Interaction and contribution breakdowns, warnings and structured executive summary.
- Saved history; review edits; JSON, text, subtitle, table and calendar exports.

Fusion comparison: the controlled diagnostic

- 24 author-designed cases; six constructed slides; top-1 correctness includes abstention.
- Temporal-only: 12/24 correct $\rightarrow$ 50.00%.
- Semantic-only: 20/24 correct $\rightarrow$ 83.33%.
- Equal-weight hybrid: 16/24 correct $\rightarrow$ 66.67%.
- Semantic-only wins this constructed diagnostic; it is not a real-meeting benchmark.

Real fusion example: coverage is not accuracy

- Real record: 1,008 passages and 34 slides; no valid slide-display timestamps.
- Temporal-only: 0 links; semantic-only: 39 links.
- Hybrid: the same 39 links, because its missing-timing fallback is semantic-only.
- Link coverage: 39/1,008 = 3.87%; semantic/hybrid agreement = 100%.
- Independent relevance labels are missing, so accuracy and F1 are unavailable.

Discussion results and the misleading-accuracy trap

- Historical reference-act cohort: precision 39.94%, recall 81.18%, F1 53.54%.
- Model accuracy: 68.62%; balanced accuracy: 73.10%.
- Always-negative baseline accuracy: 77.72%, but positive-class F1 = 0.
- High recall supports review suggestions; low precision means false positives.
- Reference-text results do not validate final decisions or raw-audio performance.

Speech and speaker evaluation: small scoped pilots

- WER measures word errors; DER measures speaker-time errors; lower is better.
- Hindi, 8 samples: Small WER 50.96%; Turbo 28.85%.
- Hinglish, 8 samples: Small 66.13%; Turbo 57.26%.
- Historical English, 2 clips: Small 61.61%; Turbo 63.91%.
- Speaker pilot: pooled DER 64.08% → 40.91%; only two held-out excerpts.

Influence research: a separate exploratory task

- GAP study: 28 meetings, 84 people and 87 comparable participant pairs.
- Compare participation features, contribution features and their combination.
- Combined Ridge: 68.93% pairwise; participation-only Ridge: 61.79%.
- Word count: 66.55%; duration: 67.86%; fixed prototype score: 55.77%.
- Outcome alignment is a proxy, not ground-truth causal influence.

Engineering validation and deployment status

- Saved earlier full-suite record: 656 Python tests; latest hosting subset: 59 passed.
- Real Docker check: 2 audio parts + 2 PPTs, 30 seconds of public English speech.
- 2 CPUs / 4 GiB limit: 33.821 seconds post-upload processing, 14 transcript segments.
- Playback, export, account isolation and volume-backed container replacement passed.
- Backend packaged for Hugging Face; public hosting and cloud latency not verified.

Missing features, necessary work and future scope

- Necessary: larger independent labels, end-to-end accuracy and deployment-storage validation.
- Necessary: reduce speaker/event errors and evaluate real reviewer usefulness.
- Not required for this scope: facial analysis, emotion scoring or a chatbot.
- Optional extensions: grounded generative notes, slide-transition capture and learned fusion.
- Do not equate an omitted convenience feature with an unimportant correctness gap.

Demonstration and conclusion

- Open a completed meeting; play an original transcript timestamp.
- Inspect a speaker, discussion passage, action and candidate decision's sources.
- Show slide links or explicit unassigned passages, then the contribution breakdown.
- Export a report and explain the metric's exact scope.
- Contribution: local, source-linked review with transparent evidence and limitations.